CHAPTER

4

SUPPLY AND ELASTICITY OF SUPPLY

Course of Action
Maximum Time = 3:15 Hrs
Maximum Questions = 85

Learning Objectives

The students will learn:

- *The meaning of supply and the various determinants of supply.*
- *The concept of supply curve and schedule.*
- *Movement along the supply curve and shift in the supply curve.*
- *Law of Supply, its assumptions, and the exceptions to the law of supply.*
- *Elasticity of Supply and the factors that affect it.*
- *Various degrees of elasticity of supply.*
- *Methods to calculate the elasticity of supply.*

Topic-1
Supply – Meaning and Determinants **Page No. 54**
Topic-2
Law of Supply and Changes in Supply **Page No. 60**
Topic-3
Elasticity of Supply **Page No. 64**

Topic-1 Supply – Meaning and Determinants

Concepts covered: Meaning of supply and stock, Concept of supply schedule and supply curve, Positive relationship between price and supply Concept of individual and market supply and determinants of supply.

Revision Notes

- **Supply:** Supply refers to the quantity of a commodity that a firm is willing and able to offer for sale at a given price during a given period of time.
- **Individual Supply**: Individual supply refers to the supply of a particular commodity at a particular price by a single firm in the market.
- **Market Supply**: Market supply refers to the supply of a particular commodity at a particular price by all the firms in the market.
- **Supply Schedule**: It is a tabular statement showing various quantities of a commodity being supplied at various levels of price, during a given period of time.
- **Individual Supply Schedule**: It is a schedule which represents different quantities of a commodity which an individual producer or seller is ready to supply at various possible prices at a given period of time.
- **Market Supply Schedule**: It is a schedule which represents the total quantity of a commodity that all producers are willing to supply at different market price per period of time. It is a horizontal summation of individual supply schedules.
- **Supply Curve**: It is a graphical representation of the supply schedule.
- **Individual Supply Curve**: Graphical presentation of individual supply schedule is called individual supply curve. It shows the different quantities of a commodity, an individual firm is willing to sell at different prices during a given time period.

Scan to know more about this topic
Determinants of supply

- **Market Supply Curve**: Market supply can be derived by horizontal summation of all individual supply curve It show the different quantities of a commodity that all the firms are willing to sell at different prices during a given time period.
- **Supply Function**: It refers to functional relationship between supply of a commodity and its determining factors.

$$Sx = f\{P_x, P_r, P_f, G, E_x, T, N_f, Gp\}$$

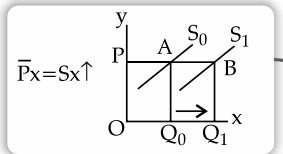
- **Factors Affecting Supply:**

(1) Price of commodity
(2) Price of related goods
(3) Price of factor inputs
(4) Goal of the firm
(5) Expected change in price

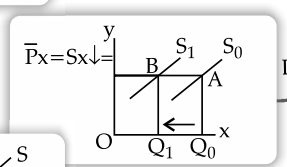
Supply and Elasticity of Supply

Change in Supply Shift in supply curve

- Increase in Supply (Rightward saved shift)

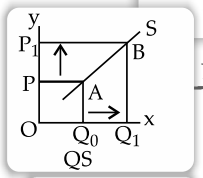


- Decrease in Supply (Leftward shift)

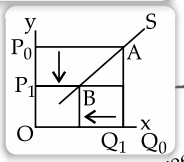


Change in quantity supply (movement along supply curve)

- Extension/Expansion of Supply (upward movements)



- Contraction of Supply (downward movement)



Elasticity of supply (ratio of percentage change in supply to percentage change in price)

- Different Degrees of e_s
  1. perfectly elastic supply, $e_s = \infty$ (horizontal supply curve)
  2. perfectly inelastic supply e_s = zero (vertical supply curve)
  3. Unitary elastic supply $e_s = 1$ (caught line supply curve originating from zero)
  4. highly Elastic supply $e_s > 1$ (supply curve intercepting OY (supply curve intercepting OX axis)
- Methods of Measuring Elasticity of Supply
  - Point Method or Geometric Method
    - Horizontal supply curve $\rightarrow e_s = \infty$
    - Vertical supply curve $\rightarrow e_s = 0$
    - Straight line supply curve passing through origin $\rightarrow e_s = 1$
    - Straight line supply curve goes through X axis $\rightarrow e_s < 1$
    - Straight line supply curve goes through Y axis $\rightarrow e_s > 1$
  - Percentage or proportionate method

$$e_s = \frac{\%\text{ Change in Quantity Supplied}}{\%\text{ Change in Price of the Commodity}}$$

$$e_s = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

$\therefore = e_s > 1$

Where, ΔQ = Change in quantity.
ΔP = Change in price
P = Initial price
Q = Initial quantity

Law of Supply

- Assumptions (these are kept constant)
  - Price of related goods
  - Future price expectations
  - Factor price/cost of in inputs
  - Technology
  - Govt. policy of Tax & Subsidy
- Exceptions
  - Agricultural goods
  - Perishable goods
  - Auction goods
  - Rare goods
  - Disposal of old stock
- Why Law of Supply Operates (causes of positive slope)
  - Profit Level
  - Entry of new firms
  - Change in stock

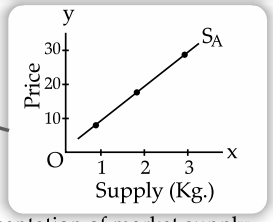
Types

- Individual supply (Supply by an individual firm)
  - Individual Supply Schedule

Tabular presentation of individual supply

P_x (₹)	S_A (kg)
10	1
20	2
30	3

  - Individual supply curve
    - Graphical presentations individual supply



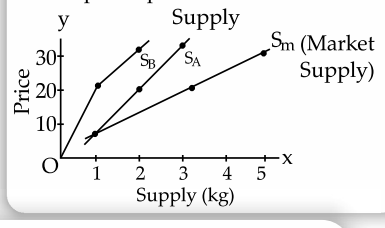
- Supply of a commodity by all producer
  - Market Supply Schedule

Tabular presentation of market supply

Px (₹)	S_A	S_B	Market Supply ($S_A + S_B$)(kg.)
10	1	0	1 + 0 = 1
20	2	1	2 + 1 = 3
30	3	2	3 + 2 = 5

  - Market Supply Curve

Graphical presentation of Market Supply



Determinants

1. Price of the commodity = $Px\uparrow = Sx\downarrow$ Px rises = Sx rises
2. Price of related Good = $Pr\uparrow = Sx\downarrow$
3. Future expectation of increase in price $Px\uparrow = Sx\downarrow$
4. Price of inputs — $C\uparrow = S\downarrow$, $C\downarrow = S\uparrow$
5. State of Technology — Improved = $S\uparrow$; Backward = $S\downarrow$
6. Government Policy — Indirect tax $\uparrow = S\downarrow$; Subsidies $\uparrow = S\uparrow$
7. Goals of firms — Expansion – $S\uparrow$
8. Number of firms$\uparrow = S\uparrow$

Trace the Mind Map
- First Level
- Second Level
- Third Level

(6) Technical know-how
(7) Number of firms in the industry
(8) Government policies

- **Stock**: It refers to total quantity of goods which is available with the sellers in the market at a particular point of time.
- **Factor Inputs:** The factors of production used in the process of production.
- **Industry:** All the producers in the market.
- **Related Goods:** The goods such as substitutes, that is can be used in place of one another, and complements, that is the goods that are used together.
- **Goal of the firm:** The firm either tries to maximise the profit or maximise the sales.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. Give the difference between supply schedule and supply curve. [OEB]

Ans. Supply schedule can be referred to as the tabular presentation which shows different quantity supplied at different prices during a given period of time. On the other hand, Supply curve is graphical presentation of supply schedule.

Q.2. Define supply. [OEB]

Ans. Supply can be defined as quantities of the commodity which producer is willing to supply at different prices in a given time period.

Q.3. What do you mean by stock? [OEB]

Ans. Stock implies the total quantity of the commodity that can be brought into the market by him at a short notice.

Q.4. What is the shape of the supply curve? [OEB]

Q.5. If the government increases the taxes, what will happen to the supply of the commodity?

Ans. The supply of the commodity will reduce, which is known as Decrease in supply, if the government increases the taxes.

Q.6. State whether the given statement is true or false with reason:

(a) "Law of supply shows inverse relationship between price and quantity supplied." [OEB]

Ans. False

Reason: The given statement is false because law of supply shows direct relationship between price and quantity supplied. As price rises, quantity supplied rises and as price falls then quantity supplied also falls.

(b) "When there is rise in taxes on the supply of goods, then it leads to fall in supply." [OEB]

Ans. True

Reason: The given statement is true because rise in taxes on the supply of the goods leads to higher cost and causes profits to fall and hence fall in supply.

(c) "When the price of inputs or factors of production rises, then it leads to rise in supply of the commodity." [OEB]

Ans. False

Reason: The given statement is false because when there is rise in price of inputs or factors of production then it leads to rise in cost of production, hence it leads to rise in supply of the commodity.

2 Marks Questions

Q.1. Leena is confused between two topics, that is intended supply and actual supply. Clarify her doubt by differentiating the two for her. [OEB]

Ans. The difference between intended supply and actual supply are:

S. No.	Basis of Distinction	Intended Supply	Actual Supply
1.	Meaning	Number of goods which the seller is willing to bring in the market.	The number of goods the seller actually brings in the market.
2.	Time Dimension	It has no time dimension.	It has to be at a particular period of time.
3.	Part of Stock	For perishable goods, stock and supply would be the same.	For durable goods, supply consists of a part of the total stock.
4.	Depends on	It depends on the production, price and procurement of goods.	It depends on the prevailing market price of the good.

[Any Two]

Q.2. How does price of factors of production affects the supply? [OEB]

Ans. When the price of factors of production increases, the cost of production also increases, which makes the product less profitable for the supplier at the prevailing market prices. This leads to decrease in the supply of the commodity. On the other hand, when the price of factors of production decreases, the cost of production decreases, making it more profitable for the supplier at the prevailing market prices. This leads them to increase in the supply of the commodity.

Q.3. The technology that the producer uses affects the supply of a commodity. Explain. [OEB]

Ans. The state of technology has a positive relation with

This question is for practice and its solution is available at the end of the chapter

the supply of the commodity. Better is the state of technology lower is the cost of production. This will result in the higher profits, and so the supply will increase. Likewise, if the technology is obsolete, the cost of production will increase, reducing the supply, as the product becomes less profitable.

Q.4. Define market supply. What will be the market supply, if Firm A sells 15 units, Firm B sells 20 units, Firm C sells 30 units of the same commodity? [OEB]

Ans. The market supply is the total quantity of a commodity produced by all the suppliers (sellers) to the market at different prices at a particular time.
Market Supply = 15 + 20 + 30 = 65 units.

Q.5. How does the number of firms in the market affect the supply? [OEB]

3 Marks Questions

Q.1. Explain any two reasons for the supply curve to be positively sloped. [ISC 2020]

OR

Explain any two reasons for the upward slope of the supply curve. [ISC 2016]

Ans. The reasons for the supply curve to be positively sloped are:

(i) Higher the price higher the profits: With the increase in the prices, the seller earns more. This leads them to increase the profit. The increasing profits leads to the seller to increase the production and hence increase the quantity supplied. Thus, establishing a positive relationship between price and quantity supplies, giving a positive slope to the supply curve.

(ii) Increase in the marginal cost: When the marginal cost increases for the supplier, they increase the price of the product. This helps them to cover the increase in the marginal cost. As, the firm is able to cover the marginal cost, the profits increases and the seller starts selling more. This gives a positive slope to the supply curve.

(iii) Higher price motivates other producers: If one firm earns higher profits, the other producers also start increasing the price. This leads to increase in supply, giving a positive slope to the supply curve.

Examiner's Comment

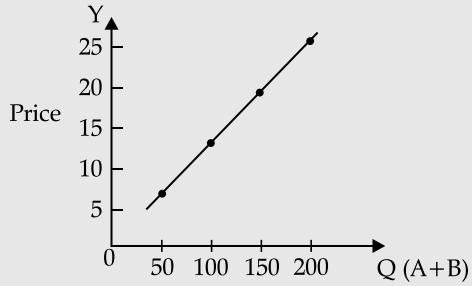
- Most of the candidates could not relate profit with supply. Many candidates wrote the positive relation between price and supply but did not write the reason behind this positive relation. A large number of candidates wrote points such as, change in input prices, use of advanced technology, etc., which are actually the reasons for shift in supply.

Answering Tip

- Students need to understand that the reasons for positive slope of supply curve is actually the explanation of law of supply. Therefore, except price, no other factors affecting supply can change.

Q.2. Derive a market supply curve from two hypothetical individual supply schedules. [ISC 2018]

Ans.



If there are only two suppliers in the market, the market supply schedule derive from the two individual supply schedules as follows:

[ISC Marking Scheme, 2018]

Examiner's Comment

- Several candidates took different prices in the supply schedule and two individual suppliers. Some candidates drew a demand schedule instead of a supply schedule. A few candidates wrote units instead of price in the schedule.

Answering Tip

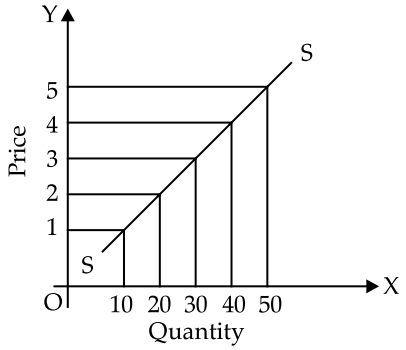
- Give sufficient practice to make a market supply schedule by first making two individual supply schedules and then drawing the market supply curve based on the market supply schedule.

Q.3. Using a diagram show whether supply of a good is directly or inversely related to its price. [OEB]

Ans. Let the following be the schedule:

This question is for practice and its solution is available at the end of the chapter

Supply schedule	
Price (₹)	**Quantity (units)**
1	10
2	20
3	30
4	40
5	50



The above table shows that more and more units of the commodity are being offered for sale as the price of the commodity is increased. As seen in the diagram, supply curve SS slope upwards from left to right, indicating direct relationship between price and quantity supplied.

Q.4. A table is given below, Complete the table. **[ISC 2013]**

Price (per unit)	Supply of A	Supply of B	Market Supply
2	7	5	?
3	?	10	18
4	9	?	25
5	11	22	?

Ans.

Price (per unit)	Supply of A	Supply of B	Market Supply
2	7	5	12
3	8	10	18
4	9	16	25
5	11	22	33

Formula used = Supply of A + Supply of B = Market Supply

Q.5. Describe in detail, the supply function. **[OEB]**

6 Marks Questions

Q.1. Discuss four determinants of supply of a commodity. **[ISC 2017]**

Ans. The determinants of supply of a commodity are:

(i) Price of the given Commodity: The most important factor determining the supply of a commodity is its price. As a general rule, Price of a commodity and its supply are **directly** related. It means, as price increases, the quantity supplied of the given commodity also rises and vice-versa. It happens because at higher prices, there are greater chances of making profit. It induces the firm to offer more for sale in the market. Supply (S) is a function of price (P) and can be expressed as: S = f (P). **The direct relationship between price and supply**, known as 'Law of Supply'. The following determinants are termed as 'other factors' or factors other than price'.

(ii) Prices of Other Goods: As resources have alternative uses, the quantity supplied of a commodity depends not only on its price, but also on the prices of other commodities. Increase in the prices of other goods makes them more profitable in comparison to the given commodity. As a result, the firm shifts its limited resources from production of the given commodity to production of other goods. For example, increase in the price of other good (say, wheat) will induce the farmer to use land for cultivation of wheat in place of the given commodity (say, rice).

(iii) Prices of Factors of Production (inputs): When the amount payable to factors of production and cost of inputs increases, the cost of production also increases. This decreases the profitability. As a result, seller reduces the supply of the commodity. On the other hand, decrease in prices of factors of production or inputs, increases the supply due to fall in cost of production and subsequent rise in profit margin. To make ice-cream, firms need various inputs like cream, sugar, machine, labour, etc. When price of one or more of these inputs rises, producing ice-creams will become less profitable and firms supply fewer ice-creams.

(iv) State of Technology: Technological changes influence the supply of a commodity. Advanced and improved technology reduces the cost of production, which raises the profit margin. It induces the seller to increase the supply. However, technological degradation or complex and out-dated technology will increase the cost of production and it will lead to decrease in supply.

(v) Government Policy (Taxation Policy): Increase in taxes raises the cost of production and, thus, reduces the supply, due to lower profit margin. On the other hand, tax concessions and subsidies increase the supply as they make it more profitable for the firms to supply goods.

This question is for practice and its solution is available at the end of the chapter

(vi) **Goals / Objectives of the firm:** Generally, supply of a commodity increases only at higher prices as it fulfills the objective of profit maximisation. However, with change in trend, some firms are willing to supply more even at those prices, which do not maximise their profits. The objective of such firms is to capture extensive markets and to enhance their status and prestige. **[Any Four]**

[ISC Marking Scheme, 2017]

Examiner's Comment

- A number of candidates wrote this answer correctly. However, in many cases determinants were not explained. In some answers, only captions were mentioned as determinants of supply. A number of candidates wrote the 'determinants of demand' instead of the 'determinants of supply'.

Answering Tips

- Students need to understand the role of profit as the guiding force behind any type of supply decisions.
- Students need to explain all the captions properly.
- Understand the determinants of demand and supply in a tabular form.

Q.2.Demand and Supply are two different concepts but are related to each other. Explain the above statement. [OEB]

Ans. The differences between Demand and Supply are:

S. No.	Basis of Distinction	Demand	Supply
(i)	Definition	Demand is the quantity of a commodity that a consumer is willing and able to buy, at each possible price during a given period of time.	Supply of a commodity refers to the quantity which a seller is prepared to sell at a given price and at a given time.
(ii)	Relation with Price	It decreases when the price increases and increases when the price decreases. Thus it has an inverse relation with price.	It decreases when the price decreases and increases when the price increases. Thus it has a positive relation with price.
(iii)	Curve	The curve is mostly downward sloping.	The curve is mostly upward sloping.
(iv)	Future Price Expectations.	If the consumer expects the price to rise in the future, the demand will increase and if they expect the price to fall in the future the demand will decrease.	If the producer expects the price to rise in the future, the supply will decrease and if they expect the price to fall in the future the supply will increase.
(v)	Schedule	The demand schedule shows the various combinations of quantity demanded at different price.	The supply schedule shows the various combinations of quantity supplied at different price.
(vi)	Law	The law of Demand states that all other factors being constant, when price of a product increases the demand of the product reduces and when price of the product decreases, demand of the product will increase.	The law of supply states that other things being remain constant, the quantity supplied varies directly with the price of the commodity. When price rises, the quantity supplied rises, and when price falls, the quantity supplied also falls.

They are related to each other as both demand and supply are the function of price of the commodity. When plotted in the same graph, the equilibrium price can be achieved. They together also help in the study of the market and the trend in the market which allows both the producers and consumers to make their respective decisions.

Q.3. Define the following terms: **[OEB]**

(a) Supply.

(b) Individual Supply.

(c) Supply Curve.

(d) Supply Function.

(e) Supply Schedule.

(f) Market Supply Schedule.

This question is for practice and its solution is available at the end of the chapter

Topic-2 Law of Supply and Changes in Supply

Concepts covered: Law Of Supply- Meaning and assumptions, Exceptions to the law of supply, Movement along the supply curve– a. Extension of supply, b. Contraction of supply, Shift in supply curve, a. Rightward shift, b. Leftward shift

Revision Notes

Scan to know more about this topic

Law of Supply

- **Law of Supply:** The law of supply depicts the relationship between price and quantity supplied of a commodity when all other determinants of supply remain constant. This law states that there is a direct relationship between price and quantity supplied of a commodity, other factors determining supply remaining constant. It means quantity supplied of a commodity increases with increase in price and decreases with decrease in price.
- **Assumptions of Law of Supply:** The Law of Supply assumes the following as constant:
  - Price of all related goods
  - Prices of inputs or factors of production
  - Techniques of production
  - Goals of the producer
  - Policies of the government
  - Expectations about the market
- **Exceptions to the Law of Supply:** (1) Rare commodities (like old coins, postal stamps, etc.), (2) Supply of labour, (3) Agricultural produce, (4) Future expectations about further changes in prices, and (5) Backward economies.
- **Change in Supply**: It is of two kinds:
  - Movement along a supply curve due to change in prices.
  - Shifting of supply curve due to change in factors other than price.
- **Movement Along the Supply Curve**: Two types of movement:
  - **Extension of Supply:** Other things being constant, when supply increases due to increase in price only. It is termed as "Extension of Supply"
  - **Contraction of Supply**: Other things being equal, when supply decreases due to decrease in price only, it is termed as "contraction of supply".
- **Shifting of Supply Curve**: Two types of shifting:
  - **Increase in Supply:** It refers to a rise in the supply of a commodity caused due to any factor other than the own price of the commodity.
    For example, when the cost of production of mobile phones decreases, the supply of phones increases.
  - **Decrease in Supply:** It refers to a fall in the supply of commodity caused due to any factor other than own price of the commodity.
    For example, if the number of firms producing clothing items decreases, the supply of clothing items also decreases.
- **Causes of Increase in Supply:**
  - Fall in the price of competing product
  - Fall in the price of factors of production
  - Improvements in technology
  - Increase in the number of firms in the market
  - Reduction in taxes or grant of subsidy
- **Causes of Decrease in Supply:**
  - Obsolescence of technology
  - Increase in the prices of substitute
  - Increase in factor prices
  - Increase in taxation or withdrawal of subsidy
  - Decrease in the number of firms in the market.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. State the difference between expansion and supply and increase in supply? [OEB]

Ans. Expansion in supply can be referred to as the rise in supply due to rise in price and increase in supply can be referred to as the rise in supply due to change in other factors except price of the commodity.

Q.2. State the difference between contraction in supply and decrease in supply? [OEB]

Ans. Contraction in supply can be referred to as fall in supply due to fall in price of the commodity, other factors remaining constant. and decrease in supply can be referred to as the fall in supply due to change in other factors except price of the commodity.

Q.3. Give the meaning of law of supply. [OEB]

Q.4. How does the perishable goods act as an exception to the law of supply? [OEB]

Ans. The supply of perishable goods does not change with the rise or fall in the price as they are regularly supplied at any given price. Thus, they act as an exception to the law of supply.

This question is for practice and its solution is available at the end of the chapter

Q.5. Will the law of supply be applicable, if the producer expects the prices to fall in the near future? [OEB]

Ans. No, the law of supply will not be applicable if the producer expects the prices to fall in the near future, as they will start to sell more at the present time, increasing the supply of the commodity.

Q.6. State whether the given statement is true or false with reason: [OEB]

"Increase in supply can be defined as when supply rises due to change in other factors such as fall in price of factor of production."

Ans. True

Reason: The given statement is true as increase in supply can be defined as change in other factors such as fall in price of inputs or factors of production.

Q.7. State whether the given statement is true or false with reason: [OEB]

"In case of movement along the supply curve, there is either rightward or leftward shift in supply curve."

Ans. False

Reason: The given statement is false because in case of movement along the supply curve, there is either expansion in supply or contraction in supply. And in case of shift in supply curve, there is either increase in supply or decrease in supply.

Q.8. State whether the given statement is true or false with reason: [OEB]

"Advancement in technique of production leads to fall in supply of the commodity".

Ans. False

Reason: The given statement is false as when the technique of production becomes advanced then it leads to rise in supply of the commodity because cost pf production falls.

Q.9. State whether the given statement is true or false with reason: [OEB]

When the wage rate increases beyond a certain limit, then number of labour hours starts falling."

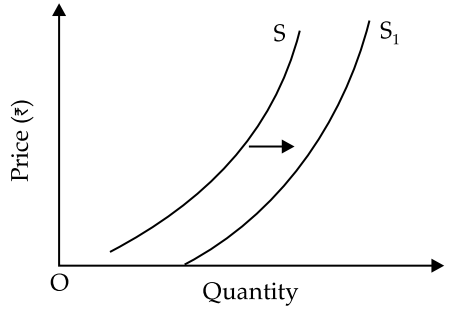
Ans. True

Reason: Beyond a certain wage rate, the labour prefers to have some relaxed hours. The workers can maintain the same standard of living by working for fewer hours at higher wage rates. As a result, beyond that wage rate, the supply of labour starts falling. As a result, the supply curve of labour is backward bending.

2 Marks Questions

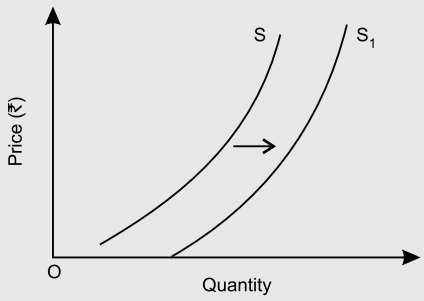
Q.1. It is claimed that Artificial Intelligence can help the manufacturing industries improve their efficiency. How will this impact the supply of the product by the manufacturing industries? Show the impact with the help of a diagram. [ISC Specimen Paper, 2023-24]

Ans. The use of artificial intelligence will lead to the increase in production as the cost of production will decrease making it more profitable, and thus the supply curve will shift to the right.



Q.2. Technical advancement leads to cost saving. With the help of a diagram, explain the effect of technical advancement on the supply curve. [ISC 2016]

Ans. If the producers use advanced technology in their process of production, the cost of production will be reduced and profit will increase. Thus, total supply will increase, so supply curve will shift to the right. Modern technology reduces cost of production.



[ISC Marking Scheme 2016]

Examiner's Comment

- 'Advanced technology leads to cost saving and therefore profit increase' was not mentioned by some of the candidates. The supply curve shifts to the right – this was not shown by many candidates.

Answering Tip

- Explain the basic reason behind the 'shift of the curve' and 'movement' along the curve.

Q.3. With the help of a diagram, explain why is there a movement along the supply curve. [OEB]

This question is for practice and its solution is available at the end of the chapter

Q.4 How is extension in supply different from the increase in supply? [OEB]

Ans. Difference between extension in supply and increase in supply are:

Basis	Extension in Supply	Increase in Supply
Meaning	When the quantity supplied increases due to increase in price, keeping other factors constant, it leads to a movement along the supply curve.	When the supply increases due to any change in the other factors, at the same price, it leads to a shift in supply curve.
Effect on Supply Curve	The movement is upward along the same supply curve.	The shift in the supply curve is rightward.
Reason	It occurs due to increase in the price of the given commodity.	It occurs due to a change in other factors like decrease in the price of inputs, decrease in taxes, improvement in technology, etc.

(Any Two)

3 Marks Questions

Q.1. Discuss any three reasons for the leftward shift of a supply curve. [ISC 2019]

Ans. The reasons for the leftward shift of the supply curve are:

(i) **Technological Changes:** When there is technological obsolescence, output is produced at higher cost. Higher cost of production decreases the supply of the commodity. The supply curve shifts to the left.

(ii) **Price of Other Goods:** When price of a substitute good increases, the quantity supplied of the concerned good decreases and there is a leftward shift of the supply curve.

(iii) **Price of Factors of Production:** If factor price increases, cost of production also increases. Accordingly, supply decreases and the supply curve shifts to the left.

(iv) **Number of Firms:** Decrease in the number of firms implies decrease in market supply and so the supply curve shifts to the left.

(v) **Increase in taxes:** Government levies tax on every unit of commodity sold or bought. Due to levy of tax, cost of production increases and by increasing in the value of tax, cost of production increases. Therefore, if taxes are increased cost of production rises and this leads to a decrease in the supply of the commodity at the same price. Thus, the supply curve shifts to the left. **[Any Three]**

Examiner's Comment

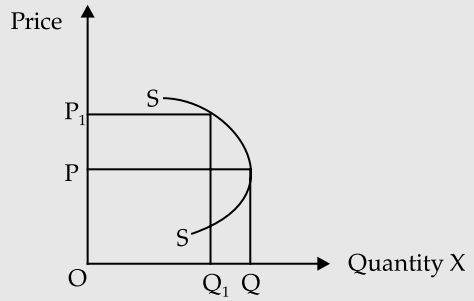
- Several candidates considered price change as one of the reasons of the leftward shift of supply curve. They got confused with the determinants of demand. A few candidates wrote the factors affecting supply instead of the factors responsible for leftward shift of the supply curve.

Answering Tips

- Explain the factors which are responsible for decrease in supply after explaining the meaning of leftward shift of the supply curve.
- Explain why demand factors cannot lead to leftward shift of supply curve.

Q.2. "The supply curve of labour is an exception to the law of supply." Justify the statement, using a diagram. [ISC 2018]

Ans. The supply of labour is an exception to the law of supply. Initially, the supply of labour follows the law of supply, that is, with an increase in wage rate, there is an increase in supply of labour. But beyond a certain wage rate, the labour prefers to have some relaxed hours. The workers can maintain the same standard of living by working for fewer hours at higher wage rates. As a result, beyond that wage rate, the supply of labour starts falling. As a result, the supply curve of labour is backward bending.



[ISC Marking Scheme, 2014]

Examiner's Comment

- Majority of the candidates did not state the reasons for fall in supply of labour due to rise in the wage rate. Several candidates were unable to draw the correct labour supply curve as an exception to the law of supply.

Answering Tips

- Explain the concept to the students that workers may work for shorter hours if wages are higher so that they can enjoy leisure time.
- Ensure that students draw the diagrams only after having understood the concept.

Q.3. Explain any two causes of a 'rightward' shift of supply curve. **[OEB]**

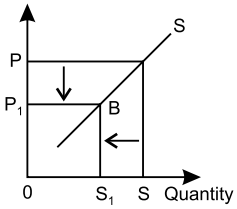
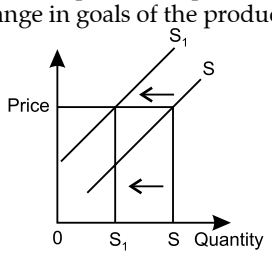
Ans. Causes of rightward shift of supply curve are:

(i) Use of improved technology: It reduces cost. Price remaining unchanged, profit rise. This leads to shift of supply curve to the right.

(ii) Fall in prices of inputs: It reduces cost. Price remaining unchanged, profits rise. This leads to shift of supply curve to the right.

Q.4. With the help of diagrams, differentiate between contraction in supply and decrease in supply. **[OEB]**

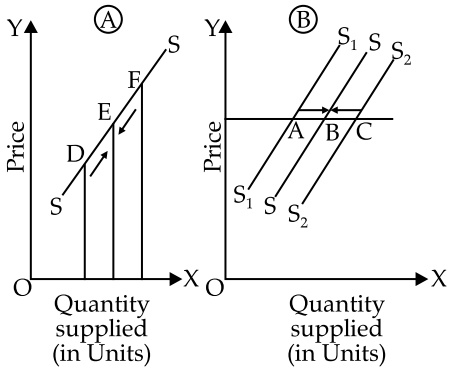
Ans.

Basis	Contraction in Supply	Decrease in Supply
Meaning	The fall in the quantity supplied due to the fall in price of the commodity is known as contraction in supply.	A decrease in supply means that producers now supply less at a given level of price of a commodity.
Effect on supply curve	The movement is downward along the same supply curve.	The shift in the supply curve is leftward.
Reason	It occurs due to decrease in the own price of the given commodity. 	It occurs due to a change in other factors like increase in the price of inputs, increase in taxes, change in goals of the producer, etc. 

Q.5. State the law of supply. "Technology plays an important role in the supply of a commodity." Explain what effect does a change in technology have on the supply? **[OEB]**

4 Marks Questions

Q.1. Study the figures given below and explain them. **[OEB]**



Ans. Change in Quantity Supplied: It refers to the change in supply due to own price of the goods only. (Refer fig. A). It means movement along the same supply curve. It is called extension when supply rises from D to E and contraction when supply falls (from F to E).

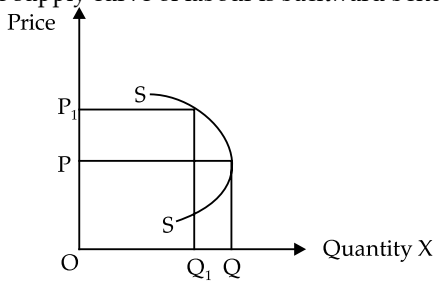
Change in Supply: It refers to change in supply due to factor other than the own price of the goods like change in prices of inputs, etc., (Refer fig. B). This leads to shift of supply curve. It is called increase in supply, when supply rises (from A to B) and decreases when supply falls from (C to B).

Q.2 State the various exceptions to the law of supply. **[OEB]**

Ans. The two exceptions to the Law of Supply are:

(i) Agricultural Produce: The supply of agricultural produce cannot be increased with increase in prices because of limitation of agricultural land and the time involved in producing a fresh crop. Also, it is more season dependent. On the other hand, most of the agricultural produce like fruits and vegetables is perishable in nature. This is why, their supply cannot be reduced with decrease in prices.

(ii) The supply of labour is an exception to the law of supply. Initially, the supply of labour follows the law of supply, that is, with an increase in wage rate, there is an increase in supply of labour. But beyond a certain wage rate, the labour prefers to have some relaxed hours. The workers can maintain the same standard of living by working for fewer hours at higher wage rates. As a result, beyond that wage rate, the supply of labour starts falling. As a result, the supply curve of labour is backward bending.

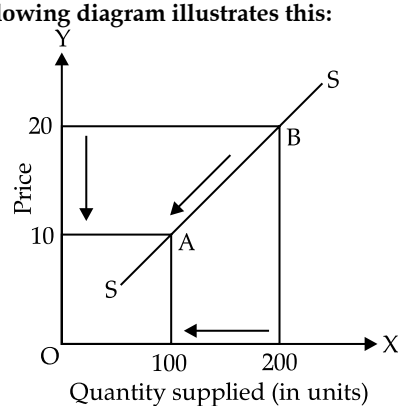


This question is for practice and its solution is available at the end of the chapter

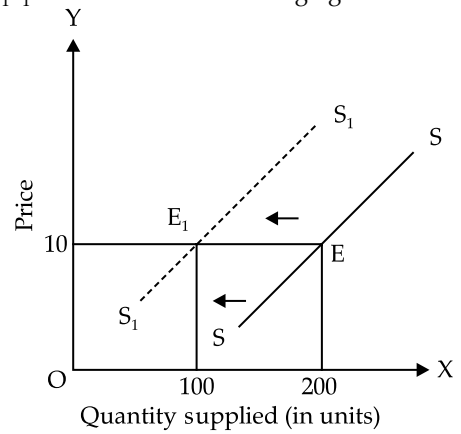
Q.3. Examine the effect of (i) fall in the own price of good X, and (ii) rise in tax rate on good X, on the supply curve. Use diagrams. [OEB]

Ans. (i) When there is a fall in the own price of good X, it will lead to contraction in supply. Contraction in supply occurs when other things being equal, quantity supplied of a commodity decreases due to fall in its own price.

Following diagram illustrates this:



(ii) Increase in taxes raises the cost of production and thus, reduces the supply, due to lower profit margin. The supply curve shifts towards the left from SS to S_1S_1 as shown in the following figure:



Q.4. Explain the law of supply. Also state the assumptions on which it is based. [OEB]

Topic-3 Elasticity of Supply

Concepts covered: Meaning of elasticity of supply, Methods of measuring elasticity of supply–a. Percentage method OR proportionate method, b. Geometric method. Factors affecting elasticity of supply.

Revision Notes

➢ **Elasticity of Supply:** Price elasticity of supply measures the degree of responsiveness of quantity supplied of a commodity to change in its price.

$$e_S = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

$$e_S = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Scan to know more about this topic

Elasticity of Supply

➢ **Measurement of Elasticity of Supply:**

- **Percentage or Proportionate Method:** This method measures the degree of responsiveness of quantity supplied of a commodity to change in its price.

The price elasticity of supply is the ratio of percentage change in quantity supplied of a commodity to percentage change in its price.

$$e_S = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

$$e_S = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Scan to know more about this topic

Factors affecting Elasticity of Supply

- **Geometric Method**: Under this method, we can conceive following three possible situations of elasticity of supply:

(a) Any straight line supply curve passing through the origin has value of elasticity equal to one $e_S = 1$.

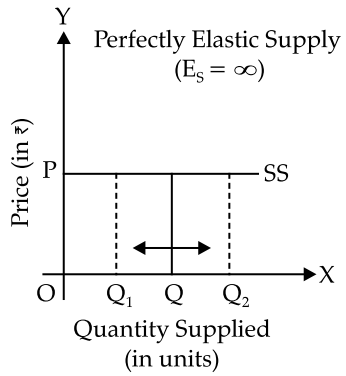
(b) If straight line supply curve goes through the X axis, it is inelastic. $e_S < 1$.

(c) If a straight line supply curve goes through the Y axis, it is elastic or highly elastic $e_S > 1$.

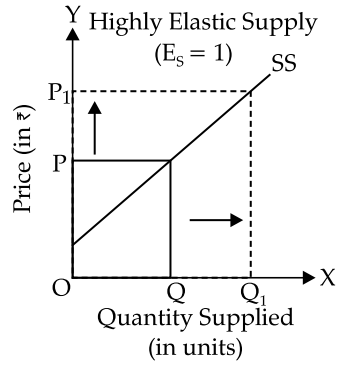
➢ **Degrees of Elasticity of Supply**:

- **Perfectly Elastic Supply** ($e_S = \alpha$): When the quantity supplied of a commodity expands or contracts to any extent without any change or with an infinitely small change in its price, the supply of the commodity is called perfectly elastic. Its supply curve is a horizontal line parallel to x-axis.

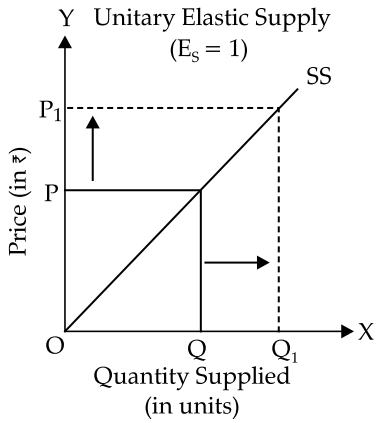
This question is for practice and its solution is available at the end of the chapter



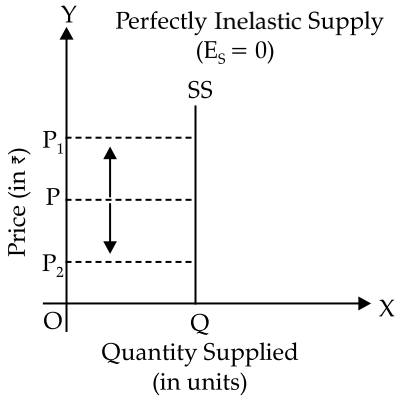
- **Greater than Unitary Elastic Supply** ($e_S > 1$): When the proportionate change in quantity supplied is more than proportionate change in price, elasticity of supply becomes greater than unity.



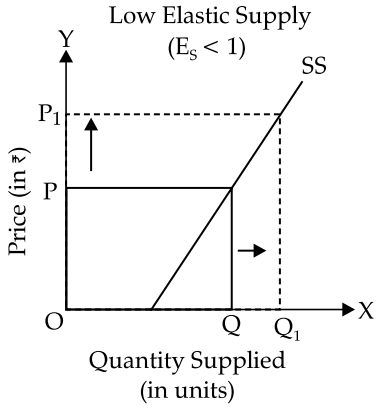
- **Unitary Elastic Supply** ($e_S = 1$): When the proportionate change in quantity supplied is equal to the proportionate change in price, the elasticity of supply becomes equal to one.



- **Perfectly Inelastic Supply** ($e_S = 0$): Supply of a commodity is said to be perfectly inelastic when the quantity supplied of a commodity does not change at all in response to change in price of the commodity. It means that the price of the commodity may increase or decrease but its quantity supplied remained the same. In such cases the price elasticity of supply is zero and supply curve is a vertical line parallel to y-axis.



- **Inelastic or Less than Unitary Elastic Supply** ($e_S < 1$): When the proportionate change in quantity supplied is less than the proportionate change in price, elasticity of supply in less than unitary.



➢ **Factors Influencing Elasticity of Supply**: (1) Nature of commodity, (2) Cost of production, (3) Estimates of future prices, (4) Natural constraints, (5) Techniques of production, (6) Nature of inputs used, and (7) Time element.

Example:

As a result of fall in the price of X from ₹30 to ₹24, its supply falls from 500 units to 450 units. Calculate its price elasticity of supply.

Ans. Given, P_1 = ₹30

P_2 = ₹24

$Q_1 = 500$

$Q_2 = 450$

$\Delta P = 24 - 30 = -6$

$\Delta Q = 450 - 500 = -50$

$$Es = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$= \frac{-50}{-6} \times \frac{30}{500}$$

$$= 0.5$$

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. State the difference between perfectly elastic supply and perfectly inelastic supply? [OEB]

Ans. The perfectly elastic supply can be referred to as when due to slight change in price, supply of the commodity changes. It's curve is parallel to x-axis. The perfectly inelastic supply can be referred to as when price changes but supply of the commodity does not change. It implies elasticity of supply is zero.

Q.2. Give difference between greater than unitary elastic supply and less than unitary elastic supply? [OEB]

Ans. The greater than unitary elastic supply can be referred to as the percentage change in quantity supplied is greater than percentage change in price. Less than unitary elastic supply can be referred to as the percentage change in supply is less than the percentage change in price.

Q.3. Give the meaning of elasticity of supply. [OEB]

Ans. Elasticity of supply can be defined as percentage change in quantity supplied of a commodity due to percentage change in factors such as price, income etc.

Q.4. If the percentage change in the price of a commodity is 20% and the percentage change in the quantity supplied is 10%, what is the price elasticity of supply? [OEB]

Ans. $$Es = \frac{\text{Percentage Change in Quantity Supplied}}{\text{Percentage Change in the Price}}$$

$$= \frac{10}{20} = 0.5$$

Q.5. Why will the agricultural good have a less elastic supply? [OEB]

Ans. The agricultural goods have a less elastic supply as the price does not affect the quantity supplied to a large extent.

Q.6. State whether the given statement is true or false with reason: [OEB]
"In a short period, elasticity of supply is elastic."

Ans. False
Reason: The given statement is false as in a short period, supply is inelastic as producer cannot change supply due to change in price in a short period whereas in long run supply is elastic.

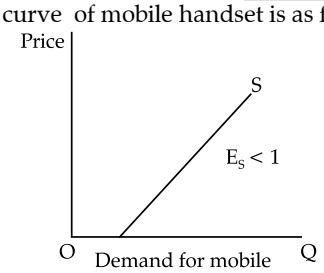
Q.7. State whether the given statement is true or false with reason:
"Availability of substitute leads to higher elasticity of supply." [OEB]

Ans. True
Reason: The given statement is true as due to availability of substitutes, when the price of one of the substitute good rises then supply of given good falls and hence supply is elastic and when price of substitute good falls then supply of given good also rises then also supply is elastic.

2 Marks Questions

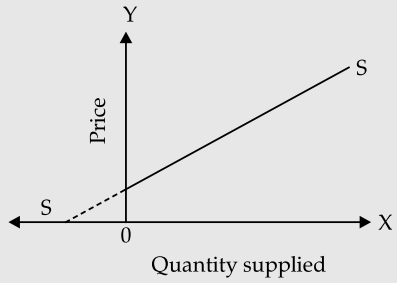
Q.1. The price of a mobile handset has risen in the market. But the dealers have not been able to increase the supply proportionately.
What will be the price elasticity of supply for the mobile handset? Draw the supply curve to indicate the type of elasticity. [ISC 2023]

Ans. When the proportionate changes in quantity supplied is less than proportionate change in price, elasticity of supply become less than unity. The supply curve of mobile handset is as follows:

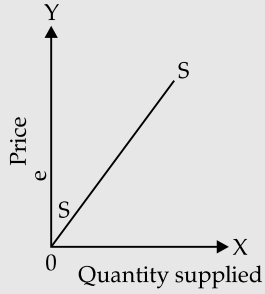


Q.2. With the help of diagrams, show when the elasticity of supply is: [ISC 2019]
(a) greater than one. (b) equal to one.

Ans. (a) Greater than one
Relatively elastic supply. $\therefore E_s > 1$



(b) equal to one
Unitary elastic supply. $\therefore E_s > 1$



[Only Diagrams]
[ISC Marking Scheme, 2019]

Examiner's Comment

- Some candidates, instead of drawing the elasticity of supply curve as linear, drew a non-linear curve. A few candidates drew diagrams to measure elasticity of demand instead of drawing the diagram to measure elasticity of supply.

Answering Tip

- Learn clearly to the students the difference between linear and non-linear supply curves and give adequate practice to draw these curves.

Q.3. How is elasticity of supply different from supply of a commodity? [ISC 2017]

Ans. Elasticity is the degree of responsiveness of change of quantity supply to the change in its price.

Supply – quantity the seller is willing to sell corresponding to the given price /It refers to the quantity of a commodity which producers are willing to produce and offer for sale at a particular price during a particular period of time.

[ISC Marking Scheme, 2017]

Examiner's Comment

- In many scripts 'degree of responsiveness' was not stated, rather, many candidates wrote proportional method instead of difference between 'elasticity of supply' and 'supply of a commodity'. For supply, the terms 'time period' and 'offer for sale' were missing.

Answering Tip

- Emphasise on the term 'degree of responsiveness' in the definition of any type of elasticity. Clearly state the difference between definition and formula.

Q.4. The following headline appeared in the newspaper– "Crop damage in Himachal Pradesh sent tomato prices soaring in Delhi." Analyse the statement with reference to the relationship between price and supply. [ISC Sem-2, 2023]

Ans. When production of tomato damaged, it means supply become scarce, their prices increase. In such case, the product can be sold at a price greater than the equilibrium price. This is the fundamental economic theory that when demand exceeds supply, prices tend to rise.

Q.5. If price of a commodity rises from ₹2 to ₹4, then supply of the commodity also rises from 20 units to 40 units. [OEB]

Ans.

Price	Supply
2	20
4	40

$\Delta P = 4 - 2 = 2$

$\Delta Q = 40 - 20 = 20$ units

$$E_s = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$E_s = \frac{20}{2} \times \frac{2}{20}$$

$$E_s = \frac{10}{10} = 1$$

3 Marks Question

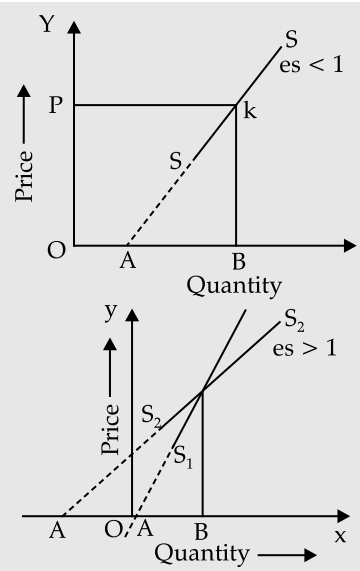
Q.1. Explain the geometric method of calculating elasticity of supply. [ISC 2020] [ISC 2015]

Ans. To measure price elasticity of supply at point k on SS curve, extend the SS curve, downward to meet the x-axis and apply the formulae .

(i) At point k on the SS curve, $\frac{AB}{OB} < 1$, therefore es < 1

(ii) At point k on S1, S1 curve, $\frac{AB}{OB} = 1$ therefore, es = 1

(iii) At point k on S2, S2 curve $\frac{AB}{OB} > 1$, therefore, es > 1



[ISC Marking Scheme, 2020]

Examiner's Comment

- Most of the candidates did not show the derivation of the price elasticity of supply. Many candidates did not label the diagrams. A few candidates got confused between > (greater) and < (less) symbols.

Answering Tip

- Students should write the formula to measure the elasticity of supply and that they draw the graph for each case. Students should draw the supply curve and then extend the curve, so that it touches X/Y axis. They should mark a point on the supply curve, draw a perpendicular from that point on X axis and Y axis and then use the formula. For the benefit of students, they need to learn by drawing different diagrams rather than using only one diagram.

Q.2. 10% increase in the price of a good raises its supply from 150 units to 180 units. Calculate its Price Elasticity of Supply. [OEB]

Ans. Given,

Percentage change in Price = 10

S1 = 150

S2 = 180

$\Delta S = S2 - S1 = 180 - 150 = 30$

$\therefore$ Percentage Change in Supply $= \frac{30}{150} \times 100 = 20\%$

$$E^s = \frac{\text{Percentage Change in supply}}{\text{Percentage Change in price}}$$

$$= \frac{20}{10}$$

$$E^s = 2$$

Q.3. At a price of ₹10 per unit the supply of a good is 80 units. When its price rises to ₹12 per unit its supply rises by 20 percent. Calculate the Price Elasticity of Supply. [OEB]

Ans. Given,

Percentage change in Supply = 20

$P_1 = ₹10$

$P_2 = ₹12$

$\Delta P = P_2 - P_2 = 12 - 10 = 2$

$\therefore$ Percentage change in Price $= \frac{2}{10} \times 100 = 20\%$

$$E^s = \frac{\text{Percentage Change in supply}}{\text{Percentage Change in price}}$$

$$= \frac{20}{10}$$

$$E^s = 1$$

Q.4. A firm supplies 10 units of a good at a price of ₹5 per unit. Price elasticity of supply is 1.25. What quantity will the firm supply at a price of ₹7 per unit? [OEB]

Ans. Given,

$P_1 = ₹5,$

$P_2 = ₹7$

$Q_1 = 10$ units,

$Q_2 = ?$

$\Delta P = 7 - 5 = 2$

$\Delta Q = Q_2 - 10$

$E_s = 1.25$

$$E_s = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$\therefore \quad 1.25 = \frac{Q_2 - 10}{2} \times \frac{5}{10}$

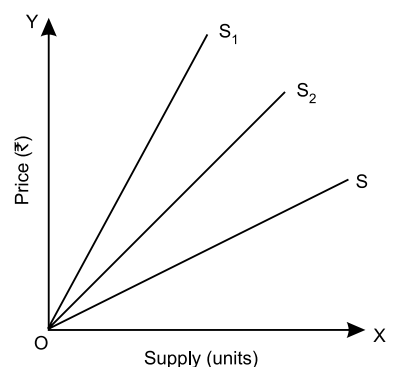
or $\quad 1.25 \times 20 = 5(Q_2 - 10)$

or $\quad 5 = Q_2 - 10$

or $\quad Q_2 = 15$ units.

Q.5. With the help of a diagram, explain what is the price elasticity associated with a straight line supply curve passing through the origin. [OEB]

Ans. Any supply curve that passes through origin will have elasticity of supply equal to one. This means that in case of any supply curve passing through origin, the percentage change in quantity supplied is equal to the percentage change in price of the commodity. In this diagram, all supply curves, S, S1 and S2 have unit elasticity of Supply.



4 Marks Questions

Q.1 Price elasticity of supply of a good is 2. A producer supplies 100 units of a good at a price of ₹20 per unit. At what price will he supply 80 units ? [OEB]

Ans. $e_s = 2$

$e_s = ?$

$$\Rightarrow 2 = \frac{20}{\Delta P} \times \frac{20}{100}$$

$$2 = \frac{20}{P-20} \times \frac{20}{100}$$

$\Rightarrow \quad 2P - 40 = -4$

$\Rightarrow \quad 2P = 36$

$P = 18$

Q.2 When price of a good rises from ₹ 8 per unit to ₹ 10 per unit, producer supplies 40 units more. Price Elasticity of Supply is 2. What is the quantity supplied before the price change? Calculate. [OEB]

Ans. Given,

Change in Quantity Supplied (ΔQ) = 40.

P = ₹ 8, P1 = ₹ 10 and $e_S = 2$

= ?

$\Delta \times P = P1 - P$

$= 10 - 8 = 2$

$$\text{Price Elasticity of Supply } (e_S) = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$\Rightarrow 2 = \frac{40}{2} \times \frac{8}{Q}$$

$$\Rightarrow 2 = \frac{160}{Q}$$

$\Rightarrow 2Q = 160$

$\Rightarrow Q = 80$ units

The Quantity Supplied before the price change was 80 units.

Q.3. A producer supplies 80 units of a good at a price of ₹10 per unit. Price elasticity of supply is 4. How much will he supply at ₹9 per unit? [OEB]

Ans. $P_1 = ₹10$,

$Q_1 = 80$ units

$P_2 = ₹9$

$Q_2 = ?$

$E_s = 4$

$\Delta P = 9 - 10 = -1$

$\Delta Q = Q_2 - 80$

$$E_s = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$\therefore \quad 4 = \frac{Q_2 - 80}{-1} \times \frac{10}{80}$$

or $\quad 4 \times -80 = 10(Q_2 - 80)$

or $\quad -32 = Q_2 - 80$

or $\quad Q_2 = 48$ units.

Q.4. When price of a good rises from ₹12 per unit to ₹15 per unit, the producer supplies 50 percent more output. What is the Price Elasticity of Supply? Comment on the shape of the supply curve as well. [OEB]

Ans. $P_1 = ₹12$

$P_2 = ₹15$

$\Delta P = 15 - 12 = 3$

$$\text{Percentage Change in Price} = \frac{3}{12} \times 100 = 25\%$$

Percentage Change in Quantity = 50%

$$E^s = \frac{\text{Percentage Change in Quantity Supplied}}{\text{Percentage Change in Price}}$$

$$= \frac{50}{25}$$

$= 2 > 1$

Thus, the commodity is highly elastic.

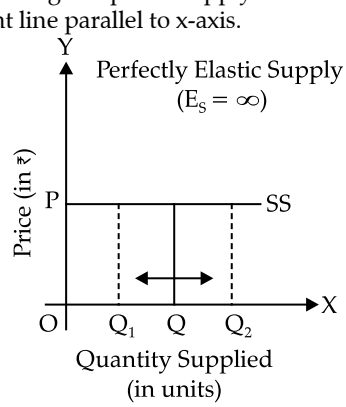
The shape of the curve will be flatter and will start from the Y-axis having a Y intercept.

6 Marks Questions

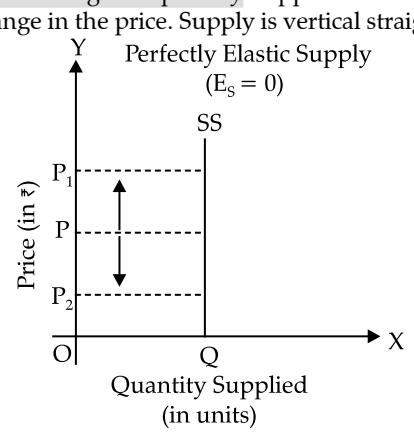
Q.1. Explain the various degrees of elasticity of supply. [OEB]

Ans. Degrees of Price Elasticity of supply are as follows:

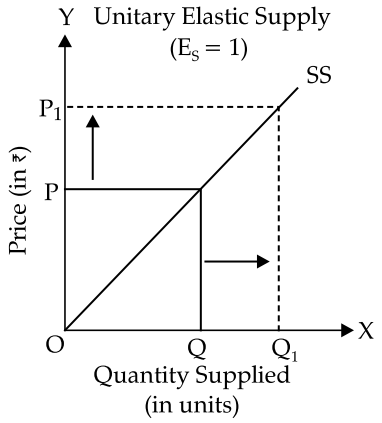
(i) Perfectly elastic supply: ($E_S = \infty$) It is a situation of infinite change in quantity supplied due to a very small change in price. Supply curve is a horizontal straight line parallel to x-axis.



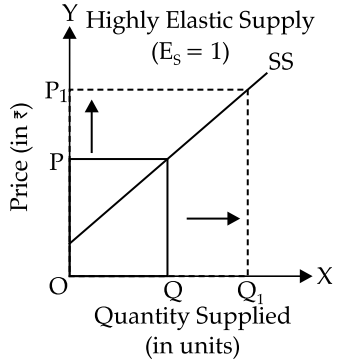
(ii) Perfectly Inelastic Supply ($E_s = 0$): It is situation of no change in quantity supplied in response to change in the price. Supply is vertical straight line.



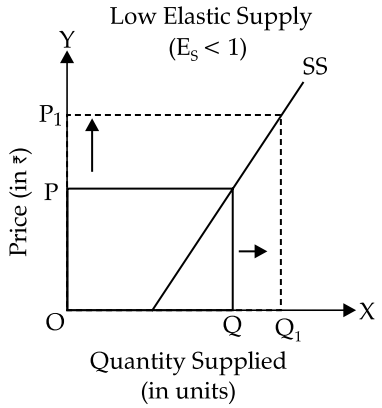
(iii) Unitary Elastic Supply ($E_s = 1$): When the percentage change in quantity supplied of a commodity is equal to the percentage change in price. Supply curve shoots from origin.



(iv) Highly or Relative or More than Unitary elastic supply (E_S > 1): When the percentage change in quantity supplied of a commodity is greater than the percentage change in price supply curve shoots from Y-axis.



(v) Highly or Relatively or Less than Unitary inelastic supply (E_S < 1): When the percentage change in quantity supplied of a commodity is less than the percentage change in price supply curve shoots from X-axis.



Q.2. Explain the factors that affect the elasticity of supply. [OEB]

Ans. Determinants of Elasticity of Supply:

(i) Nature of the inputs used: The elasticity of supply depends on the nature of inputs used for the production of commodity. If factors of production are those which are commonly used (and therefore easily available), supply of the commodity will be elastic. On the other hand, if specialized factors are used (which are not easily available), supply will be less elastic.

(ii) Natural Constraints: The elasticity of supply is also influenced by the natural constraints in the production of a commodity. If we wish to produce more teakwood, it will take years of plantation before it becomes usable. Supply of teakwood will therefore be less elastic.

(iii) Risk taking: The elasticity of supply depends on the willingness of entrepreneurs to take risk. If entrepreneurs are willing to take risk, the supply will be more elastic. On the other, hand, if entrepreneurs hesitate to take risk, the supply will be inelastic.

(iv) Nature of the commodity: Perishable goods are relatively less elastic in supply than durable goods because it is difficult to store the perishable goods.

(v) Cost of production: Elasticity of supply is also influenced by cost of production. If production is subject to law of increasing costs, then supply of such goods will be inelastic.

(vi) Time Factor: Longer the time period, greater will be the elasticity of supply. Because, over a long period of time, more and more factors are easily available, and their input can be changed to increase (or decrease) output of the commodity.

(vii) Technique of production: If the technique is complex and needs large stock of capital, then the supply of the commodity will be less elastic, because production cannot be easily increased. On the other hand, goods involving simple technique of production will have more elastic supply.

Q.3. Explain any two methods of calculating the elasticity of supply. [OEB]

Ans. Measurement of Elasticity of supply:

(i) Percentage or proportionate method: The percentage method measures the price elasticity of supply as the percentage change in the quantity supplied to the percentage change in the price of the commodity.

The percentage method measures price elasticity of supply by dividing the percentage change in quantity supplied by percentage change in price of commodity.

(a) The elasticity of supply is unity, greater than unity and less than unity.

(b) Supply is unity if change in supply is proportionate to the change in price.

(c) Supply is greater than unity when change in supply is more than proportionate change in price.

(d) The supply is less than unity if change in supply is less than proportionate change in price.

(e) The coefficient of price elasticity of supply is always positive because change in price brings a change in supply in same direction.

It is measured using the following formula:

$$\text{Elasticity of Supply} = \frac{\text{Percentage change in the quantity Supply of the commodity}}{\text{Percentage change in the price of the commodity}}$$

$$E_s = \frac{\frac{\Delta Q}{Q}\times 100}{\frac{\Delta P}{P}\times 100} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \frac{\Delta Q}{\Delta P}\times\frac{P}{Q}$$

(ii) Geometric or Point Method: According to geometric method, elasticity is measured at a given point on the supply curve. This method is also known as 'Arc Method' or 'Point Method'. Under this method, we can conceive following three possible situations of elasticity of supply:

(a) Any straight line supply curve passing through the origin has value of elasticity equal to one (ES = 1).

(b) If a straight line supply curve goes through the X axis, it is inelastic (ES < 1).

(c) If a straight line supply curve goes through the Y axis, it is elastic or highly elastic (ES > 1).

OBJECTIVE TYPE QUESTIONS

[A] Multiple Choice Questions

Q.1. GST has been imposed at the rate of 5% both on edible oil and a few spices. In this situation, the supply curve of edible oil will:
[ISC Specimen Paper, 2023-24]

(A) shift leftward. **(B)** shift rightward.
(C) not change its position.
(D) slope downward.

Ans. Option (A) is correct.

Explanation: Because, With the increase in the tax rates the cost of production of the firm will increase and hence the profit margins will fall. This will induce the firm to reduce the supply and supply curve will shift towards left.

Q.2. Which one of the following is NOT a ceteris paribus assumption of the Law of Supply? [ISC 2023]

(A) Indirect taxes and subsidies on the firm's products remain constant.
(B) Technology used by the firm in production remains constant.
(C) Prices of the factors of production paid by the firm remain constant.
(D) Price of the commodity produced by the firm remains constant.

Ans. Option (D) is correct.

Explanation: From the given options, only "price of the commodity produced by the firm remain constant" is not a correct assumption of law of supply. Ceteris paribus assumptions of law of supply are: The price of other commodities, State of technology, factor cost and taxation laws remain the same.

Q.3. When supply of a commodity increases by 24% following the rise in price by 8%, supply curve will be: [ISC Specimen Paper, 2022-23]

(A) positively sloped with positive intercept.
(B) positively sloped with negative intercept.
(C) horizontal. **(D)** parallel to price axis.

Ans. Option (A) is correct.

Explanation: Supply increases by 24% due to an 8% increase in price means that the price elasticity of supply is highly elastic. This makes the supply curve to be positively sloped and having a positive intercept as it starts from the Y-axis.

Q.4. Which of the following does not result in shift of supply curve of a good? [ISC Semester-I, 2022]

(A) Price of inputs
(B) Price of good
(C) Goods and service tax
(D) Subsidy

Ans. Option (B) is correct.

Explanation: Price of the good leads to the movement along the supply curve. If the price increases, it causes an upward movement and if the price decreases, it causes a downward movement along the same supply curve.

Q.5. Stock of a commodity is also called:
[ISC Semester-I, 2022]

(A) Intended Supply **(B)** Actual Supply
(C) Market Supply **(D)** Individual Supply

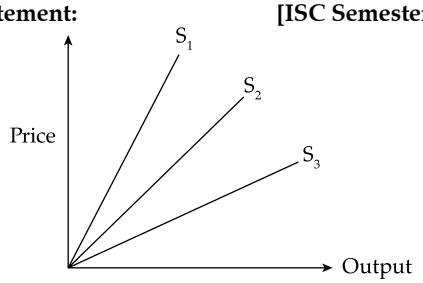
Q. 6. Movement along the supply curve occurs due to:
[ISC Semester-I, 2022]

(A) Increase in own price of the commodity.
(B) Decrease in own price of the commodity.
(C) Change in factors other than own price of the commodity.
(D) Increase and decrease in own price of the commodity.

Ans. Option (D) is correct.

Explanation: When there is an increase in the own price of the commodity, it is termed as extension of supply leading to an upward movement along the supply curve. Likewise, when there is a decrease in the own price of the commodity, it is called the contraction of supply leading to a downward movement along the supply curve. Thus, increase and decrease in own price of the commodity leads to a movement along the supply curve. Change in factors other than own price of the commodity leads to the shift in the supply curve.

Q.7. Based on the given diagram, choose the correct statement: [ISC Semester-I, 2022]



(A) S_1 is highly elastic, S_2 is unitary elastic and S_3 is highly inelastic.
(B) S_2 is relatively elastic, S_2 is unitary elastic and S_3 is highly inelastic.
(C) Both S_1 and S_2 are unitary elastic and S_1 is highly elastic.
(D) S_1, S_2 and S_3 are unitary elastic.

Ans. Option (D) is correct.

Explanation: When the percentage change in quantity supplied of a commodity is equal to the

This question is for practice and its solution is available at the end of the chapter

percentage change in price, it is termed as unitary elastic supply. The supply curve in this case shoots from origin. In the figure, all the three curves start from the origin, thus all the three are unitary elastic, irrespective of their slope.

Q. 8. If the government provides tax concession and subsidies, the supply curve will: **[ISC Semester-I, 2022]**

(A) Shift leftwards.
(B) Shift rightwards.
(C) Remain constant.
(D) Demand curve will shift.

Ans. Option (B) is correct.

Explanation: When the government provides tax concession and subsidies, it makes it more profitable for the seller to sell more. Thus the supply will increase, without the increase in the price, leading to a rightward shift in the supply curve.

Q.9. Supply is elastic if: **[ISC Semester-I, 2022]**

(A) A 10% change in price causes a larger % change in quantity supplied.
(B) The good in question is a normal good.
(C) The slope of the supply curve is positive.
(D) A 1% change in price causes a smaller % change in quantity supplied.

Ans. Option (A) is correct.

Explanation: When the percentage change in quantity supplied of a commodity is greater than the percentage change in price, it is called a highly elastic supply. Only in option (a), when the price changes by 10%, the quantity supplied increases by larger percentage, thus, it is an elastic supply.

Q.10. The shape of supply curve of perishable commodity will be: **[ISC Semester-I, 2022]**

(A) Vertical. **(B)** Horizontal.
(C) Upward sloping. **(D)** Downward sloping.

Ans. Option (C) is correct.

Explanation: The supply curve of perishable commodity will be vertical as the elasticity of supply for such commodity is perfectly elastic.

Q.11. A supply of labour decreases with increase in wage rate because: **[ISC Semester-I, 2022]**

(A) Labour starts substituting leisure for wages.
(B) Labour starts substituting wages for leisure.
(C) Demand for labour changes.
(D) Supply of labour is dependent on demand.

Ans. Option (B) is correct.

Explanation: Initially, when the wage rate is high or increasing then the supply of labour tends to increase. But later the supply of labour decreases with the increase in the wage rate because the labour will substitute leisure for wage and work less, giving a backward bend to the supply curve.

Q.12. Adoption of cost saving technological progress will: **[ISC Semester-I, 2022]**

(A) Increase the supply.
(B) Reduce cost of production.
(C) Reduce cost and increase supply.
(D) Supply will not change.

Ans. Option (C) is correct.

Explanation: When a cost saving technology is used, it will reduce the cost of production. This will make the production of goods more profitable. As a result the suppliers will supply more, increasing the supply at prevailing price.

Q.13. An increase in wages for labour will lead to: **[ISC Semester-I, 2022]**

(A) Upward movement along the supply curve.
(B) Downward movement along the supply curve.
(C) Supply curve will shift towards left.
(D) Supply curve will shift towards right.

Ans. Option (C) is correct.

Explanation: When the wages increase for labour, it will lead to a rise in the cost of production. This leads to the good being less profitable at the prevailing price rates. As a result the supplier will decrease the supply, shifting the supply curve towards the left.

[B] Assertion Reason Questions

Directions: In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.1. Assertion (A): The supply curve of a good shift to the right when prices of other goods fall.
Reason (R): When price of other good falls, it becomes less profitable to produce them in place of the given good. **[OEB]**

Ans. Option (A) is correct.

Explanation: As when the price of other goods falls, it becomes less profitable to produce them. This leads to the decrease in the supply of the good as less producers will supply it being more profitable, leading to a rightward shift of the supply curve.

Q.2. Assertion (A): With increase in the subsidy, the supply curve shifts to the left.
Reason (R): The government subsidy makes the goods more profitable for the producers to produce them. **[OEB]**

Ans. Option (D) is correct.

Explanation: When the government increases the subsidy in the goods, the producers can produce more of the goods as it will cost less to them. This will lead to the increase in supply leading to Rightward the supply curve.

Q.3. Assertion (A): When the price of the goods falls, the supply curve shifts to the right.
Reason (R): When price of the goods falls, the producers sell less as there is a positive relation between price and quantity supplied. **[OEB]**

Ans. Option (D) is correct.

Explanation: When there is a change in the price of goods, there is a movement along the supply curve and not shift in the supply curve and not shift, making assertion false. When the price of goods falls, the producers sell less as it becomes less profitable for them to sell the goods, decreasing the supply. This is because price and quantity supplied are positively related. This makes reason true.

Q.4. Assertion (A): The supply of food grains is inelastic.
Reason (R): The supply of food grain is not responsive to the price of the food grain. **[OEB]**

Ans. Option (A) is correct.

Explanation: As food grains is a necessity and have very less to the responsiveness to the price changes, the supply of food grain is inelastic.

Q.5. Assertion (A): The percentage change in quantity supplied is 20% and the percentage change in price is 20%. Thus, the elasticity of supply is 1.

Reason (R): The commodity supplied has unitary elastic supply and its supply curve will be upward rising. **[OEB]**

Ans. Option (B) is correct.

Explanation: As the change in the quantity supplied and price is same, thus the commodity has a unitary elastic supply. Thus, the Assertion is a reason for the Reason and not either way.

Case Based Questions

[A] Multiple Choice Questions

I. Read the following passage and answer Questions that's follows on the basis of the same: [OEB]

Agriculture provides livelihood to almost three fourth of the population of India. Indian agriculture is highly dependent on spatial and temporal distribution of rainfall. Climate extremes such as drought and flood affect agriculture severely. An account of impact of climate extremes viz. drought and flood, on Indian food grain production has been presented in this paper. There are temporal fluctuations in food grain production and area under the food grain. In secular terms, both of them increased up to mid-eighties.

After mid-eighties there is decline in the area of food grain while maintaining an increase in production of food grain suggesting the improvement in agricultural technology and policy. There is more temporal fluctuation in the production of food grain than the area under food grain. The analysis reveals that impact of drought on Indian agriculture is more than that of flood. Rabi food grain production depicts better adaptability to drought than Kharif food grain production mostly due to better access to irrigation infrastructure. Among the various food crops analysed all except jowar can effectively face flood events. Wheat and jowar perform relatively better during drought events.

Rice is most sensitive crop to the extreme climate events. Since rice is staple food in the sub-continent, management of rice productions against climate extremes needs special attention for food security and sustainability.

Source: www.researchgate.net

(a) What has caused the increase in the supply of food grains.

(A) Improvement of technology.
(B) Increase in production.
(C) Better agricultural policy.
(D) All of the above.

Ans. Option (D) is correct.

Explanation: The supply of food grains has increased due to improvement of technology, increase in productivity, better agricultural policies of the government, climatic conditions etc.

(b) What other things affect the supply of goods?

(A) Price of the commodity.
(B) Income of the consumers.
(C) Substitute goods price change.
(D) All of the above.

Ans. Option (D) is correct.

Explanation: The other factors that affect the supply of a commodity are: Price of related goods, Price of factor inputs, Goal of the firm, Expected change in price, Technical know-how, Number of firms in the industry and Government Policies

(c) The _____ of jowar and wheat is relatively better during monsoon.

(A) Demand **(B)** Supply
(C) Both (A) and (B) **(D)** Neither (A) nor (B)

Ans. Option (B) is correct.

Explanation: Among the various food crops analysed all except jowar can effectively face flood events. Wheat and jowar perform relatively better during drought events.

Q.4. _____ affects the supply of agricultural goods in India.

(A) Climate **(B)** Technology
(C) Government Policy **(D)** Number of Supply

Ans. Option (A) is correct.

Explanation: Climate extremes such as drought and flood affect agriculture severely.

Q.5. _____ is the main reason for the increase in the supply of food grains.

(A) Providing Government Subsidies
(B) Decreasing Taxes
(C) Improving Technologies
(D) Increasing Productivity

Ans. Option (C) is correct.

Explanation: As per the passage, "After mid-eighties there is decline in the area of food grain while maintaining an increase in production of food grain suggesting the improvement in agricultural technology and policy."

[B] Subjective Based Questions

I. The responsiveness of change of the quantity supplied with respect to its price is called elasticity of supply. Elasticity can be categorised into various types with respect to the degree of elasticity. When the quantity is less responsive to the price change it is termed as inelastic and when a little change in the price changes the quantity supplied many folds, it is termed as elastic supply. If there is no change in the price with the change in the quantity supplied, it is termed as perfectly elastic supply and if the price changes irrespective of the change in supply, it is termed as perfectly inelastic supply. In case if the price and quantity supplied of the product is equal then it is unitary elastic. Thus elasticity measures the responsiveness of the quantity and price with respect to the change in one of the factors. **[OEB]**

(a) Which type of elasticity is responsiveness of change in quantity supplied to the change in price of the commodity. Why?

Ans. Supply because elasticity of demand can be defined as the percentage change in demand divided by the percentage change in price. whereas elasticity of supply can be defined as degree of responsiveness in supply due to change in price.

(b) What are the various degrees of elasticity of supply?

Ans. Various types of degrees of elasticity of supply are perfectly elastic supply, perfectly inelastic supply,

higher elastic supply, inelastic supply, unitary elastic supply.

(c) Which degree of elasticity is it when the supply of the commodity is not responsive to the change in its price?

Ans. Perfectly inelastic supply because there is no change in supply in response to change in price.

(d) State whether the given statement is true or false: In perfectly elastic supply, price is responsive to the change in supply.

Ans. False, because price does not response to the change in supply.

(e) State with reason that elasticity of supply is affected by various factors.

Ans. Elasticity of supply is affected by nature of the commodity, cost of production, techniques of production, etc.

III. Elasticity of supply is a measure of a producer's ability to cope effectively with changes in supply. A number of factors can affect it.

Availability of resources is a factor. If a company depends on an increasingly scarce resource to produce its product, it may be unable to step up production when demand increases. Moreover, the resource will become increasingly expensive, forcing a corresponding increase in the producer's price or decrease in its production, or both.

Technology innovation is a factor in many industries. More efficient production reduces costs and allows for larger production numbers at lower prices.

The number of competitors is a factor. An increase in the number of suppliers makes the price of a product or service more elastic. If one supplier can't meet demand, others will rush to fill the gap.

Flexibility is a big factor. If a resource becomes scarce, can another resource be substituted? Can production be ramped up quickly in response to greater demand? Efficient producers can respond more quickly to increased demand. **[OEB]**

(a) What does the elasticity of supply measure?

Ans. Elasticity of supply measures, the producer's ability to cope up with the changes in supply .

(b) When is the producer unable to increase production even if the demand rises?

Ans. When the producer is dealing with the supply of a product whose resources are scarce, it will not be able to increase the supply with increase in the demand.

(c) State whether the given statement is true or false: Better the technology innovation, more will be the supply of the product.

Ans. True, because innovation in technology will lead to shift in supply curve towards right.

(d) What are the different degrees of elasticity of supply in which supply curve is either vertical or horizontal?

Ans. The different degrees of elasticity of supply are:

Perfectly elastic supply: When elasticity of supply is equal to infinity, then perfectly elastic supply.

Perfectly inelastic supply: When elasticity of supply is equal to zero, then perfectly inelastic supply.

(e) If percentage change in supply is 40% due to percentage change in price by 20%, then calculate price elasticity of supply?

Ans. E_s = (% change in supply)/(% change in price)

$E_s = 40/20$

$E_s = 2$

Solutions for Practice Questions (Topic-1)

1 Mark Questions

4. The supply curve is upward sloping.

2 Marks Questions

5. There is a positive impact of the changes in number of firms in the supply. More the firms, more will be the supply and lesser is the firm, less will be the supply at the given price level.

3 Marks Questions

5. Supply Function: It refers to functional relationship between supply of a commodity and its determining factors.

$$S_x = f\{P_x, P_y, P_f, G, E_x, T, N_f, Gp\}$$

Where, P_x = Price of commodity

P_y = Price of related goods

P_f = Price of factor inputs

G = Goal of the Firm

E_x = Expected change in price

T = Technical know-how

N_f = Number of firms in the industry

G_p = Government Policies

6 Marks Questions

3. (a) Supply refers to the quantity of a commodity that a firm is willing and able to offer for sale at a given price during a given period of time.

(b) Individual supply refers to the supply of a particular commodity at a particular price by a single firm in the market.

(c) Supply curve is the graphical representation of different levels of quantity supplied at different price levels.

(d) Supply function refers to the functional relationship between supply of a commodity and its determinants.

(e) Supply Schedule is the tabular statement which represents various quantities of a commodity being supplied at various levels of price during a given period of time.

(f) Market supply schedule is the supply schedule which represents the total quantity supplied by all the producers in the market at various price levels per period of time.

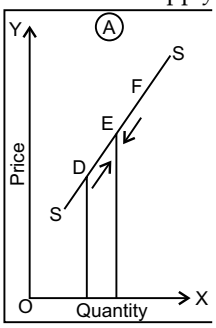
Solutions for Practice Questions (Topic-2)

1 Mark Questions

3. Law of supply can be referred to as the direct relationship between price and quantity supplied. As price rises, supply rises and as price falls, supply also falls.

2 Marks Questions

3. Change in Quantity Supplied is the situation when there is movement along the supply curve. It refers to the change in supply due to own price of the goods only. It means movement along the same supply curve. It is called extension when supply rises from D to E and contraction when supply falls (from F to E).



3 Marks Questions

5. Law of supply states that, other things being equal, quantity supplied increases with increase in price and decreases with decrease in price of a commodity.

   If the producers use advanced technology in their process of production, the cost of production will be reduced and profit will increase. Thus, total supply will increase, so supply curve will shift to the right. Modern technology reduces cost of production.

4 Marks Questions

4. **Law of Supply** states that, other things being equal, quantity supplied increases with increase in price and decreases with decrease in price of a commodity.

   The law can be explained with the help of following schedule and diagram:

   Supply Schedule

Price (₹)	entity (units)
1	10
2	20
3	30
4	40
5	50

The above table shows that more and more units of the commodity are being offered for sale as the price of the commodity is increased. As seen in the diagram, supply curve SS slope upwards from left to right, indicating direct relationship between price and quantity supplied.

Assumptions of Law of Supply: The Law of Supply assumes the following as constant:

(i) Price of all related goods
(ii) Prices of inputs or factors of production
(iii) Technique of production
(iv) Goals of the producer
(v) Policies of the government
(vi) Expectations about the market

Solutions for Objective Type Questions

[A] Multiple Choice Questions

5. **Option (A) is correct.**

Explanation: Supply is the quantity of goods sold in a particular time at a particular price. Stock refers to the total quantity of goods which is available with the seller in the market at a particular point of time which represents intended supply.

REFLECTION

(a) Can you explain the various factors that affect the supply of a commodity?
(b) What are the factors that lead to the movement along the supply curve and shift in the supply curve?
(c) Can you state the law of supply, its assumptions, and exceptions?
(d) Can you calculate the elasticity of supply and state its degree?